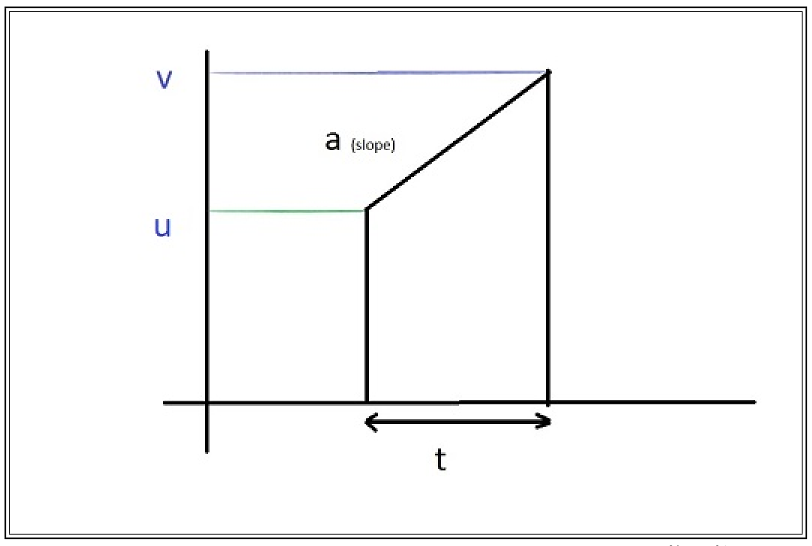
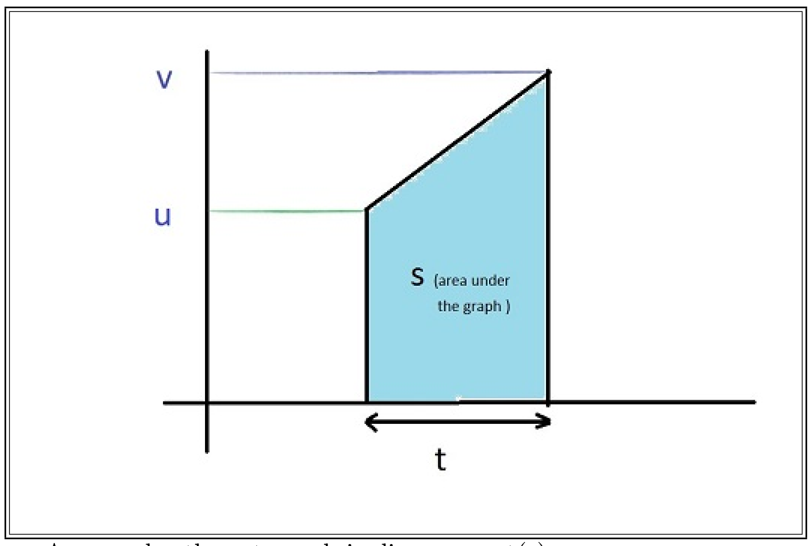
1. Derivation from v-t graph { $v = u + at$ }



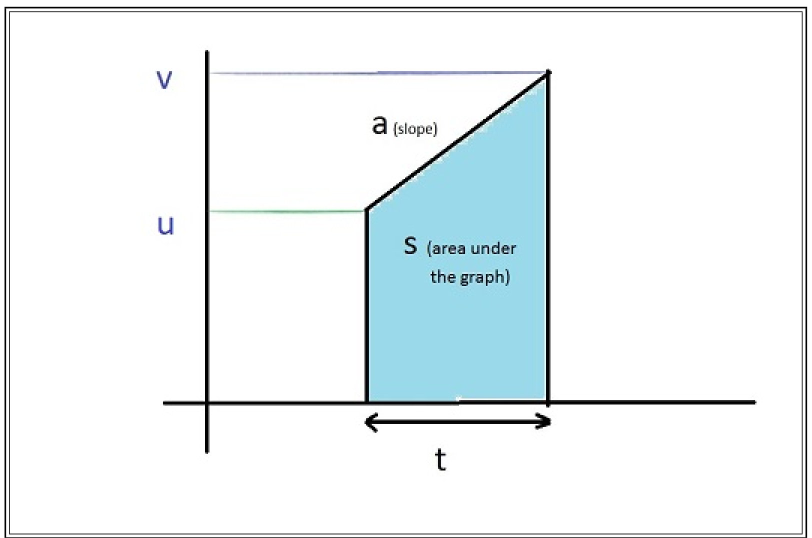
We define the slope of v-t graph as a, so it gives $a = \frac{v-u}{t}$
Manipulating the form of this equation, we get $v = u + at$, which is newton's first equation of motion in scalar form.

2. Derivation from v-t graph { $s = ut + \frac{1}{2}at^2$ }



Area under the v-t graph is dispacement(s)
Area of a trapezium is sum of parallel sides X distance between them
$\Rightarrow s = \frac{v+u}{2}t$
Substituting, the previously derived newton's first equation of motion in scalar form
$\Rightarrow s = \frac{u+at+u}{2}t$
OR
$s = ut + \frac{1}{2}at^2$

3. Derivation from v-t graph { $v^2 - u^2 = 2as$ }



In deriving the newton's third equation by graph, two values are taken from the graph
$a = \frac{v-u}{t}$ and $s = \frac{v+u}{2}t$
Multiplying both the equations,
$as = \frac{v^2-u^2}{2}$
OR
$v^2 - u^2 = 2as$

4. Result from Area Under the v-t graph, $s = v_{av}t = \frac{v+u}{2}t$

5. Manipulating $v = u + at$ as $u = v - at$ and substituting in $s = \frac{v+u}{2}t$ gives
$s = \frac{v-at+v}{2}t \Rightarrow s = vt - \frac{1}{2}at^2$